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JupyterLab Python 3 (ipykernel)

```
[6]: #Online Shoping Cart(E-Commerce)

print("-----Online Shoping Cart-----")
class ShoppingCart:
    def __init__(self):
        self.total_price = 0
    def add_item(self, price):
        self.total_price += price
        print(f"Item added! Price: {price}")
    def remove_item(self, price):
        if price <= self.total_price:
            self.total_price -= price
            print(f"Item removed! Price: {price}")
        else:
            print("Error: Cart value cannot be negative!")
    def checkout(self):
        print(f"Total Amount to pay: {self.total_price}")

my_cart = ShoppingCart()
my_cart.add_item(1200)
my_cart.add_item(500)
my_cart.remove_item(200)
my_cart.checkout()

-----Online Shoping Cart-----
Item added! Price: 1200
Item added! Price: 500
Item removed! Price: 200
Total Amount to pay: 1500
```

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JupyterLab Python 3 (ipykernel)

```
[9]: #Triangle Area Calculator(Geometry)

print("\n-----Calculate Area of a Triangle-----")
class Triangle:
    def __init__(self, base, height):
        self.base = base
        self.height = height
    def calculate_area(self):
        return 0.5 * self.base * self.height
t = Triangle(10, 15)
area = t.calculate_area()
print(f"Area of Triangle: {area}")

-----Calculate Area of a Triangle-----
Area of Triangle: 75.0

[13]: #Smart Home Device(Fan Control)

print("\n-----Smart Fan Controller-----")
class SmartFan:
    def __init__(self):
        self.speed = 0
    def speed_up(self):
        if self.speed < 5:
            self.speed += 1
            print(f"Speed increased to: {self.speed}")
        else:
            print("Fan is already at maximum speed (5)!")
    def speed_down(self):
        if self.speed > 0:
            self.speed -= 1
            print(f"Speed decreased to: {self.speed}")
```

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JupyterLab Python 3 (ipykernel)

Area of Triangle: 75.0

```
[13]: #Smart Home Device(Fan Control)

print("\n-----Smart Fan Controller-----")
class SmartFan:
    def __init__(self):
        self.speed = 0
    def speed_up(self):
        if self.speed < 5:
            self.speed += 1
            print(f"Speed increased to: {self.speed}")
        else:
            print("Fan is already at maximum speed (5)!")
    def speed_down(self):
        if self.speed > 0:
            self.speed -= 1
            print(f"Speed decreased to: {self.speed}")
        else:
            print("Fan is already turned OFF!")
    def status(self):
        print(f"Current Fan Speed: {self.speed}")
fan = SmartFan()
fan.speed_up()
fan.speed_up()
fan.speed_down()
fan.status()

-----Smart Fan Controller-----
Speed increased to: 1
Speed increased to: 2
Speed decreased to: 1
Current Fan Speed: 1
```

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JupyterLab Python 3 (ipykernel)

Current Fan Speed: 1

```
[14]: #Employee Salary Calculator(HR System)

print("\n-----Employee Salary Calculator-----")
class EmployeeSalary:
    def __init__(self, name, basic, hra, da):
        self.name = name
        self.basic = basic
        self.hra = hra
        self.da = da
    def calculate_net_salary(self):
        total_salary = self.basic + self.hra + self.da
        print(f"Employee Name: {self.name}")
        print(f"Basic pay: {self.basic}")
        print(f"Net Salary Breakdown: {total_salary}")
emp = EmployeeSalary("Vijay", 25000, 5000, 3000)
emp.calculate_net_salary()
```

-----Employee Salary Calculator-----
Employee Name: Vijay
Basic pay: 25000
Net Salary Breakdown: 33000

[]: